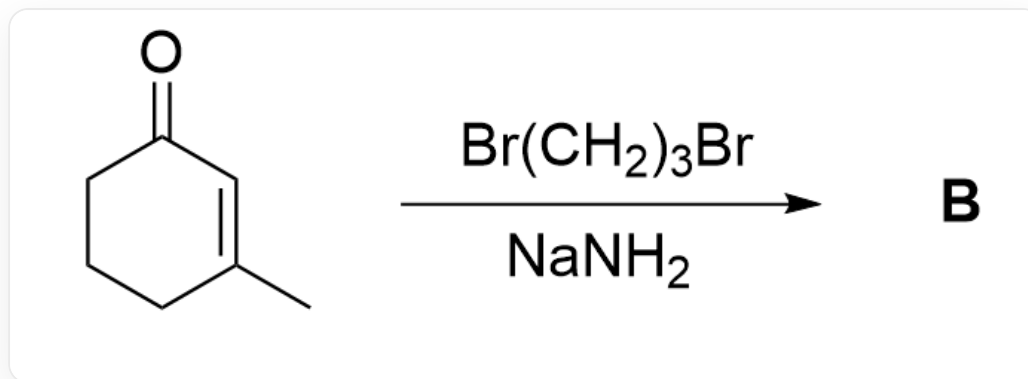
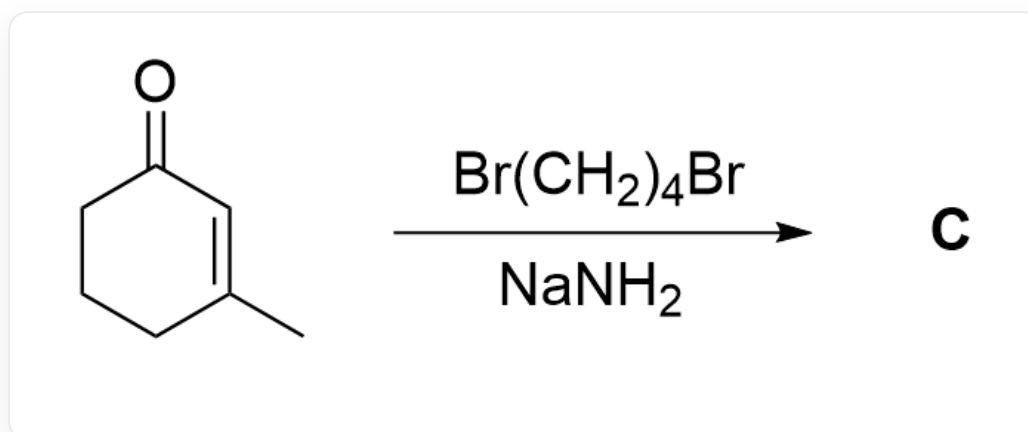


Question



O=C1C=C(C)CCC1>BrCCCBBr.[Na]N>[B], **B** is the product

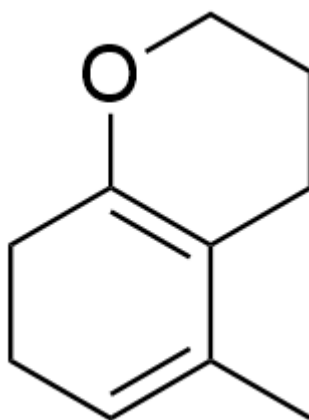


O=C1C=C(C)CCC1>BrCCCCBr.[Na]N>[C], **C** is the product

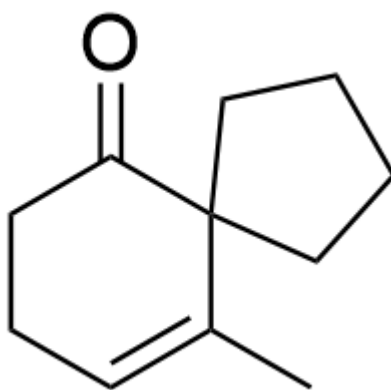
Given that the products **B** and **C** obtained in the liquid ammonia solution of sodium amide are slightly different, and without considering enantiomerism, try to give the structural formulas of products **B** and **C** respectively.

A. All other options are incorrect

B.

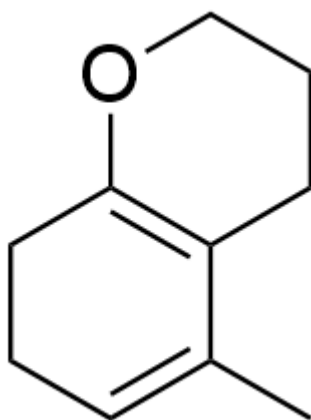


CC1=CCCC2=C1CCCO2

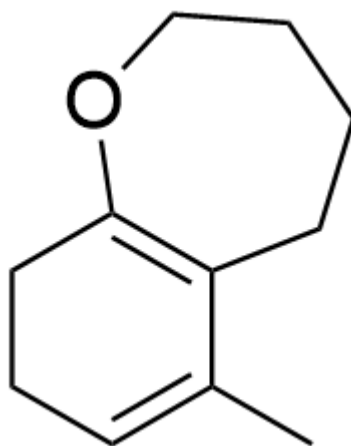


O=C1C2(CCCC2)C(C)=CCC1

C.

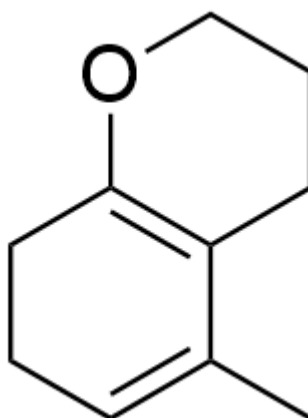


CC1=CCCC2=C1CCCO2

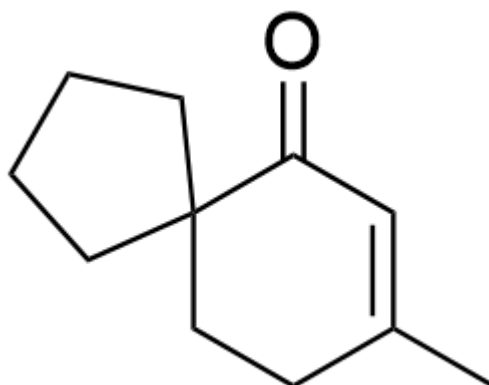


CC1=CCCC2=C1CCCCO2

D.

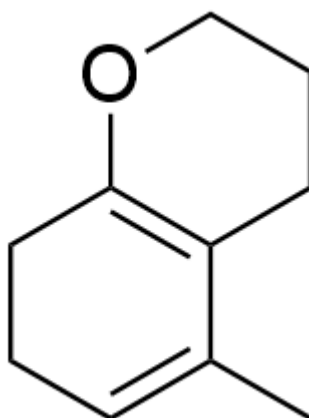


CC1=CCCC2=C1CCCO2

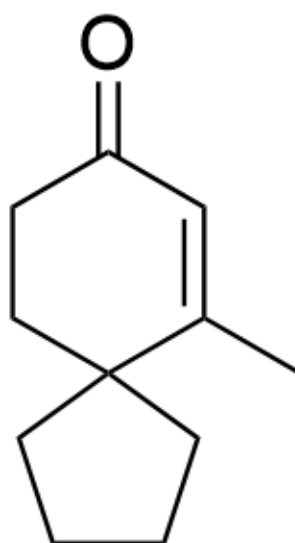


O=C1C=C(C)CCC12CCCC2

E.

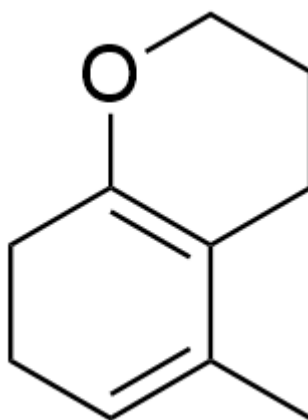


CC1=CCCC2=C1CCCO2

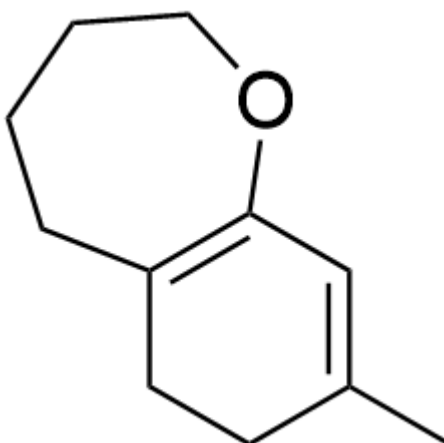


O=C1C=C(C)C2(CCCC2)CC1

F.

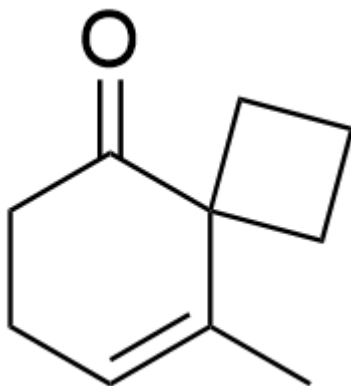


CC1=CCCC2=C1CCCO2

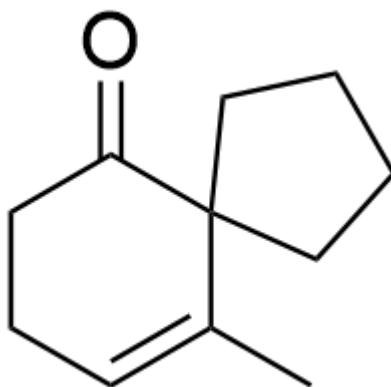


CC1=CC(OCCCC2)=C2CC1

G.

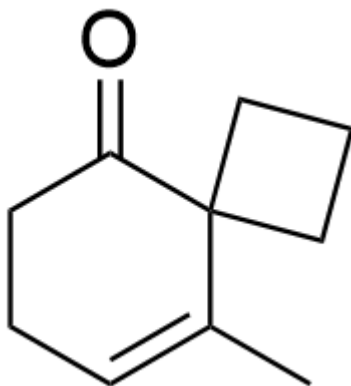


O=C1C2(CCC2)C(C)=CCC1

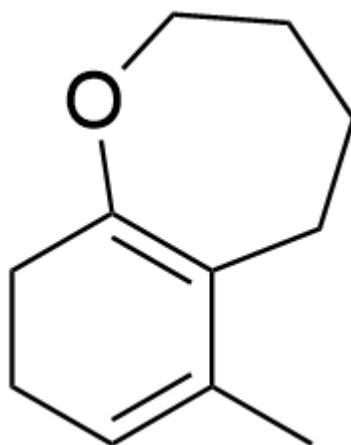


O=C1C2(CCCC2)C(C)=CCC1

H.

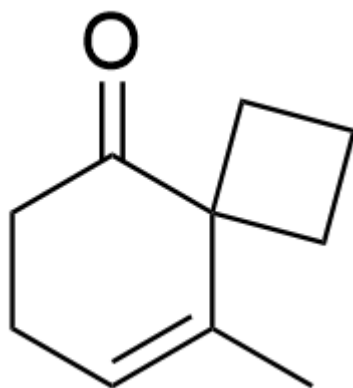


O=C1C2(CCC2)C(C)=CCC1

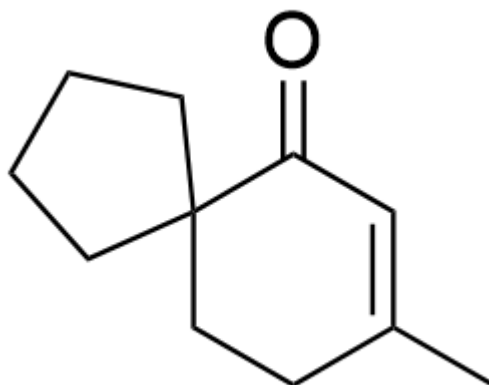


CC1=CCCC2=C1CCCCO2

I.

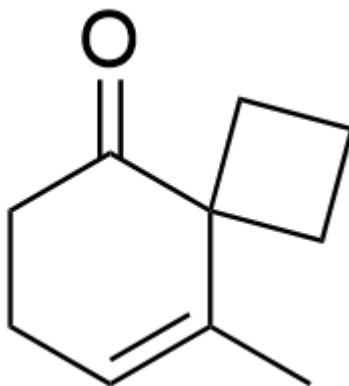


O=C1C2(CCC2)C(C)=CCC1

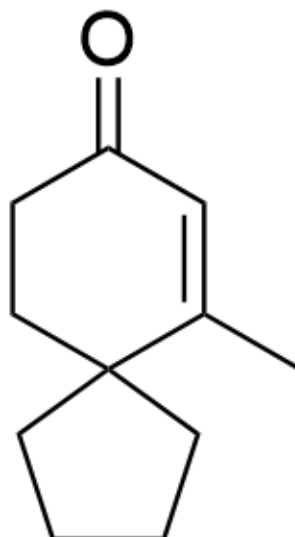


O=C1C=C(C)CCC12CCCC2

J.

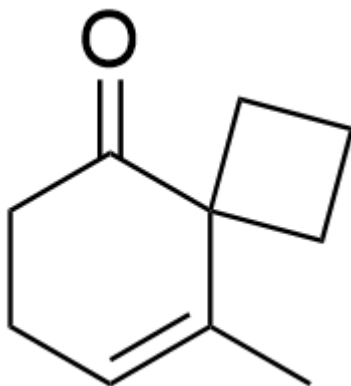


O=C1C2(CCC2)C(C)=CCC1

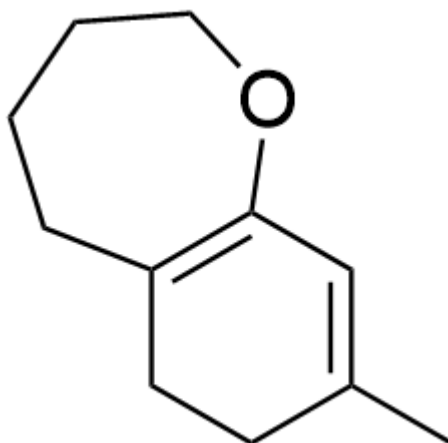


O=C1C=C(C)C2(CCCC2)CC1

K.

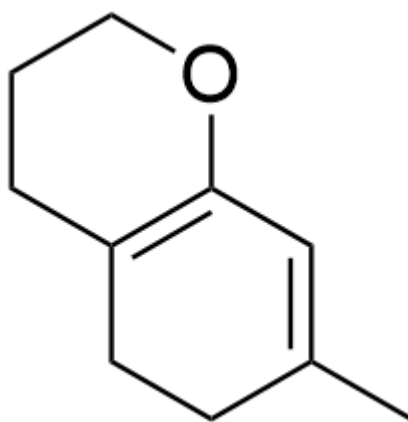


O=C1C2(CCC2)C(C)=CCC1

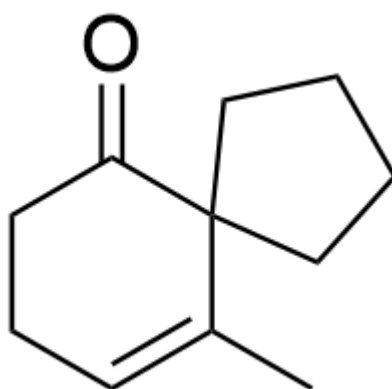


CC1=CC(OCCCC2)=C2CC1

L.

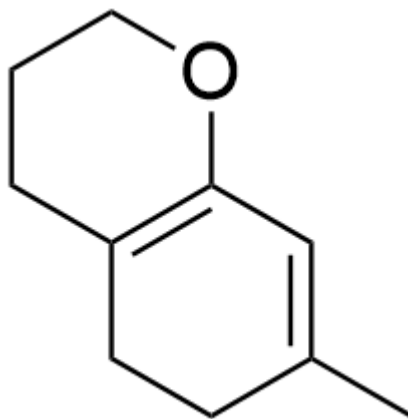


CC1=CC(OCCC2)=C2CC1

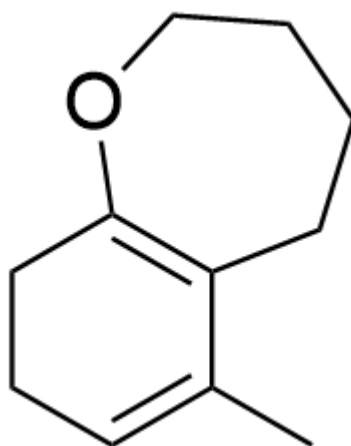


O=C1C2(CCCC2)C(C)=CCC1

M.

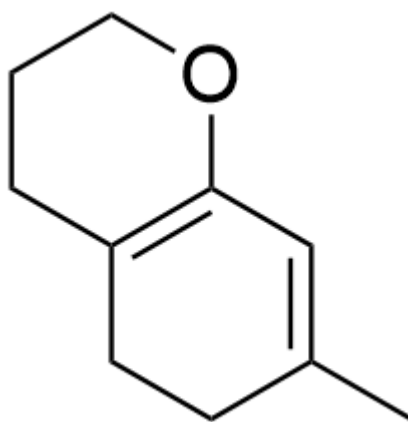


CC1=CC(OCCC2)=C2CC1

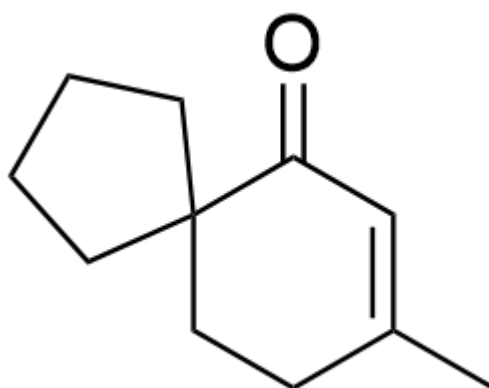


CC1=CCCC2=C1CCCCO2

N.

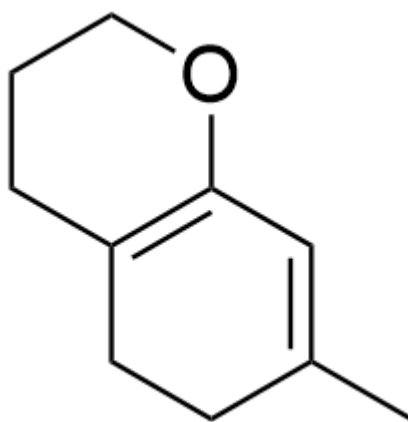


CC1=CC(OCCC2)=C2CC1

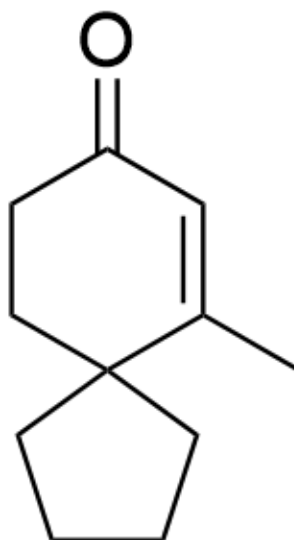


O=C1C=C(C)CCC12CCCC2

O.

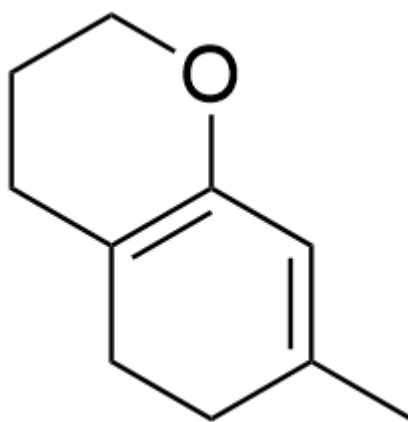


CC1=CC(OCCC2)=C2CC1

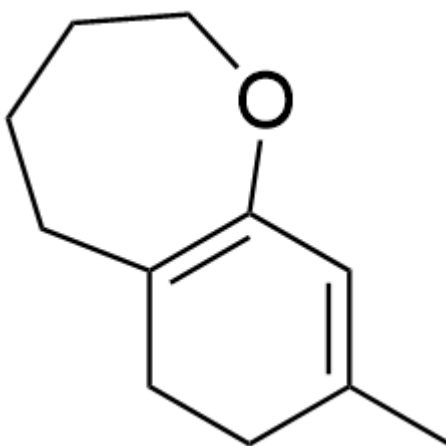


O=C1C=C(C)C2(CCCC2)CC1

P.



CC1=CC(OCCC2)=C2CC1



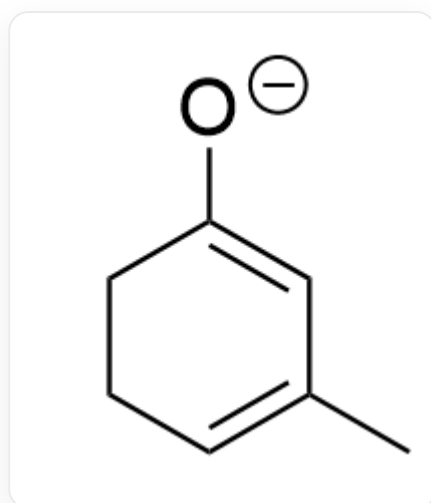
CC1=CC(OCCCC2)=C2CC1

Answer

Correct Answer: **B**

Detailed Explanation

First, in the liquid ammonia system, the thermodynamically stable linear conjugated anion intermediate **1** tends to form preferentially.



[O-]C1=CC(C)=CCC1

CHECKPOINT

1 PTS

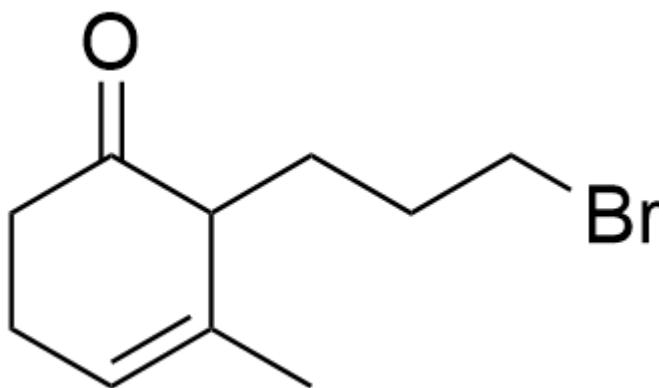
Anion intermediate **1**: [O-]C1=CC(C)=CCC1

For the reaction to generate products **B** and **C**, since the carbanion is more stable at the α position of the carbonyl group, it first reacts with the corresponding haloalkanes to obtain intermediate **2** and intermediate **3** respectively.

CHECKPOINT

1 PTS

Since the carbanion is more stable at the α position of the carbonyl group, it first reacts with the corresponding haloalkanes to obtain intermediate **2** and intermediate **3**

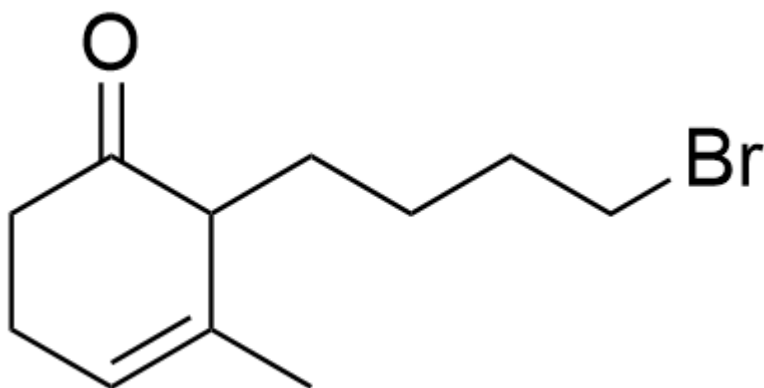


Intermediate **2**: O=C1C(CCCBr)C(C)=CCC1

CHECKPOINT

1 PTS

Intermediate **2**: O=C1C(CCCBr)C(C)=CCC1



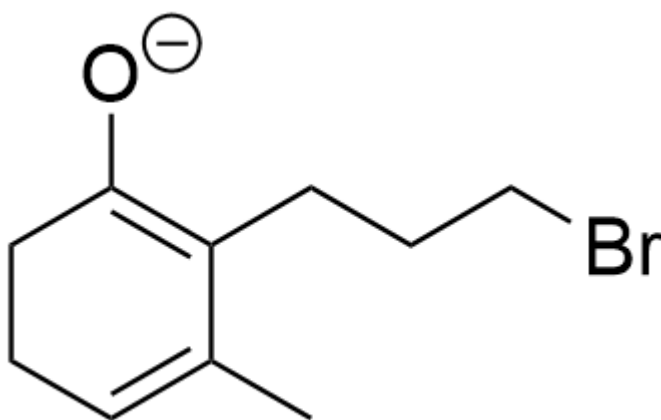
Intermediate **3**: O=C1C(CCCCBBr)C(C)=CCC1

CHECKPOINT

1 PTS

Intermediate **3**: O=C1C(CCCCBBr)C(C)=CCC1

Then, a proton is abstracted by a base to obtain intermediate **4** and intermediate **5** respectively.

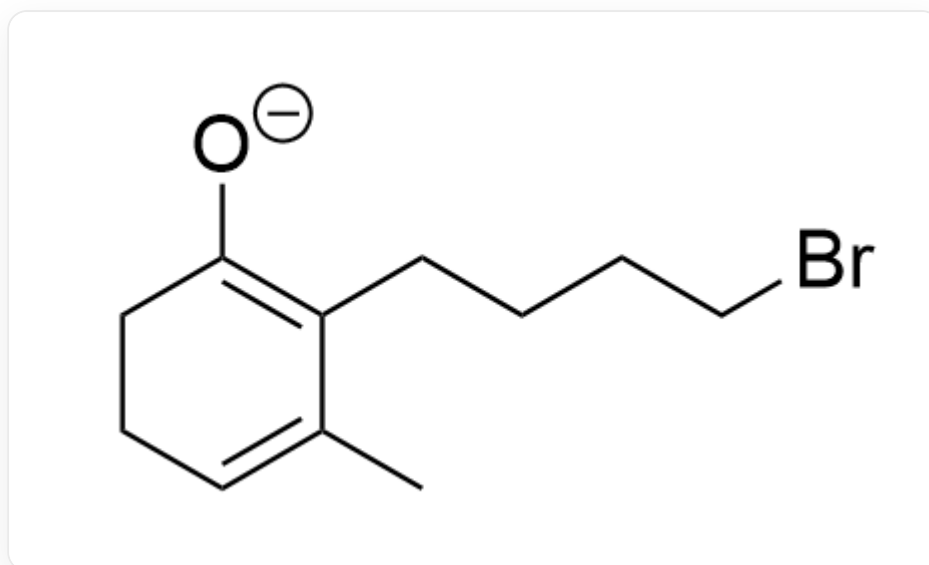


Intermediate **4**: [O-]C1=C(CCCCBBr)C(C)=CCC1

CHECKPOINT

1 PTS

Intermediate 4: [O-]C1=C(CCCBr)C(C)=CCC1



Intermediate 5: [O-]C1=C(CCCBr)C(C)=CCC1

CHECKPOINT

1 PTS

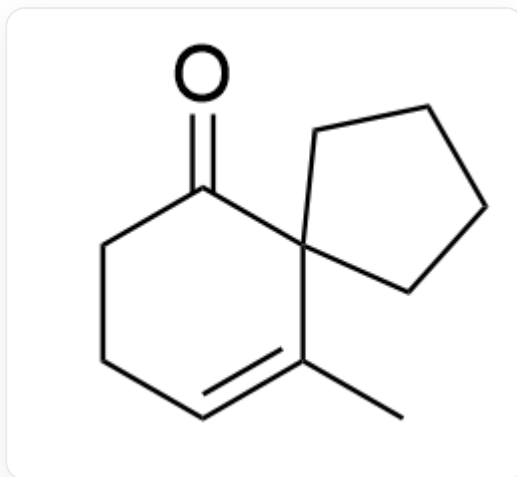
Intermediate 5: [O-]C1=C(CCCBr)C(C)=CCC1

For intermediate 5, since the carbon-end nucleophilicity is stronger than the oxygen-end, the product **C** containing a five-membered ring is preferentially obtained.

CHECKPOINT

1 PTS

For intermediate **5**, since the carbon-end nucleophilicity is stronger than the oxygen-end, the product **C** containing a five-membered ring is preferentially obtained



Product **C**: O=C1C2(CCCC2)C(C)=CCC1

CHECKPOINT

1 PTS

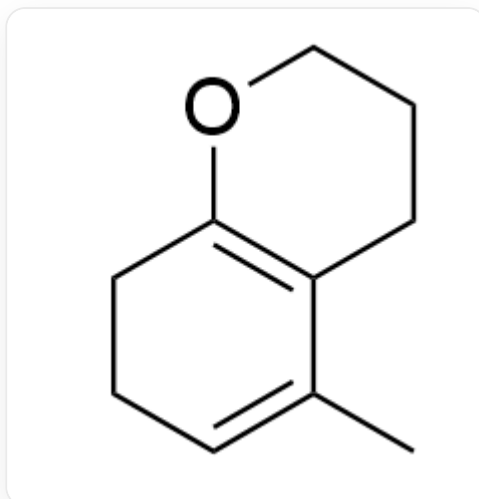
Product **C**: O=C1C2(CCCC2)C(C)=CCC1

For intermediate **4**, since the four-membered ring formed by carbon-end nucleophilic attack is unstable, it tends to use oxygen-end nucleophilic attack to form product **B** containing two six-membered rings.

CHECKPOINT

1 PTS

For intermediate **4**, since the four-membered ring formed by carbon-end nucleophilic attack is unstable, it tends to use oxygen-end nucleophilic attack to form product **B** containing two six-membered rings



Product **B**: CC1=CCCC2=C1CCCO2

CHECKPOINT

1 PTS

Product **B**: CC1=CCCC2=C1CCCO2